

Uniform sampling of a sinusoid

Background

Uniform sampling reads a continuous signal at $t[n] = n/fs$, where fs is the sampling frequency in Hz. The sampling interval is $T = 1/fs$. Here $x(t) = A \sin(2 \pi f t + \phi)$, with phase ϕ in radians.

Your task

Evaluate the signal at N uniformly spaced instants starting at zero. Return the times and values, then display the provided continuous-reference and sample plot. Change fs from 20 Hz to 40 Hz and observe the denser samples.

Function contract and assumptions

`sample_sine(A, f, phi, fs, N)` returns (t, x) , two one-dimensional arrays of length N . Assume $fs > 0$, N is a positive integer, and $f \geq 0$. The final time is $(N-1)/fs$; do not add an endpoint.

Example / correctness check

For $A=2$, $f=1$, $\phi=0$, $fs=4$, $N=4$: $t = [0, 0.25, 0.5, 0.75]$ and $x = [0, 2, 0, -2]$, up to floating-point roundoff.

Exam instructions

Complete only the TODO functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF pages 4-6.